

A Glass Box for the Clinic: A Transparent, Dual-LLM Framework for Explainable Medical Report Analysis

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Abstract—Use of Large Language Models (LLMs) in medicine is not transparent to prevent a “black box” problem wherein evidence-based clinical decisions are impossible. In response to this, we suggest a transparent dual-LLM structure, processing medical reports with Explainable AI (XAI) and Retrieval-Augmented Generation (RAG). Our XAI-RAG system is an auditable system of query processing, evidence retrieval, and response generation. In a direct comparison, the API-based GROQ-GPT model was better in its lexical performance (ROUGE-L: 0.239) compared to the local T5-Flan model (0.196), while T5-Flan achieved higher semantic similarity. The framework was tested against a ground-truth set of five pairs of question-answer based on clinical texts. This work exhibits a practical glass box approach, which makes it possible to develop trustworthy and explicable AI systems to improve clinical processes.

Index Terms—Explainable AI, Retrieval-Augmented Generation, Large Language Models, Medical Informatics, Clinical AI, Model Interpretability

I. INTRODUCTION

A. The “Black Box” Dilemma in Clinical AI

The medical diagnostics, treatment planning and administrative process are being revolutionized with the AI interpreting complex data like CT scans and forecasting the disease before it shows up. Nevertheless, this makes deep learning and LLM models obscure to clinical use, in which all decisions should be evidence-traceable. Any model that generates outcomes without an explanation compromises the standards of the medical profession and the trust of the clinicians in it [1]. This non-transparency is essential when it comes to the problem of LLM hallucinations, the convincing yet untrue outputs, which can be life-threatening in medicine. Research conducted recently has shown that clinically tuned LLMs can mitigate such risks, basing the response on facts in the data [2].

B. The Promise of Explainable RAG (XAI-RAG)

Retrieval-Augmented Generation (RAG) is an enhancement of factual accuracy based on a combination of an LLM reasoning capacity and authenticated external data. The system will first retrieve trusted information before responding, thereby providing evidence-based outputs, which is essential to clinical reliability-criteria [3] of evidence-based clinical systems. Although RAG does reduce hallucinations, it does not in itself justify its reasoning. The proposed Explainable RAG (XAI-RAG) is an extension of RAG, where the Explainable AI (XAI) techniques are integrated with all phases, i.e., query interpretation, evidence retrieval, and answer generation, to explain the manner in which every decision is taken. It answers such important questions: How is the query processed? Why did a particular evidence have to be selected? How was it utilized in the shaping of the answer?

C. A Novel Dual-LLM Comparative Framework

The study presents a comparative paradigm that evaluates two different LLM areas of clinical tasks. A 780M-parameter locally deployable model named T5-Flan (google/flan-t5-large) improves data privacy and provides predictable and hardware-controlled costs. On the other hand, GROQ-GPT (openai/gpt-oss-120b)—a 120B-parameter, open-weight model running on the Groq high-speed LPU infrastructure—provides state of the art reasoning through API access. The two-side arrangement allows conducting a comparative analysis of on-premise and cloud deployment, trade-offs in performance, scalability, cost, and data sovereignty, as a viable foundation of clinical AI model choice.

D. Summary of Core Contributions

This article describes research on the design, implementation, and evaluation of a transparent AI on clinical document analysis. The essence of the contributions is as follows:

- 1) The complete, end-to-end XAI-RAG framework design and implementation. This “glass box” system gives multi-level explanations of query processing, evidence retrieval, and response generation, allowing the whole reasoning process of AI to be auditable and transparent.
- 2) A data-driven, rigorous comparison of medical RAG embedding models. The work questions the consensus view that domain-specific models are superior, showing quantitatively that a current general-purpose language model (BAAI/bge-m3) can do better than a biomedical-specific model (BioBERT) at this retrieval task.
- 3) The first direct comparative experiment of a local T5-Flan model versus high-performance openai/gpt-oss-120b model on the Groq platform on a clinical NLP task.
- 4) A multi-metric assessment framework using XAI tools with complementary evaluative metrics (ROUGE-L, Semantic Similarity, Exact Match).

II. BACKGROUND AND RELATED WORK

A. Retrieval-Augmented Generation: Grounding LLMs in Factual Evidence

Retrieval-Augmented Generation (RAG) is a method that helps to increase the accuracy and reliability of LLM by bringing external sources of knowledge to the generation process [4]. In contrast to traditional models that use only fixed parametric memory to produce responses, RAG retrieves factual evidence and then produces responses anchored in known context before verification, and minimizes hallucinations.

RAG works in three phases, including: 1) *Indexing*, where documents are chunked and transformed into the form of vector embeddings, stored in a searchable database; 2) *Retrieval*, where user queries are embedded and compared through Approximate Nearest Neighbor (ANN) search to get the most relevant evidence; and 3) *Augmented Generation*, where retrieved chunks and the query are combined to prompt the LLM to generate evidence-based output. This is what is referred to as prompt augmentation making sure that responses are based on authoritative data.

B. Explainable AI for Natural Language Processing: Deconstructing LIME and SHAP

The black-box mechanisms of decision-making used by AI systems require clarification to create trust and ensure effectiveness in AI systems [6]. The model-agnostic tools of LIME and SHAP are popular in NLP with respect to interpretability, which are embedded in the XAI-RAG framework to explain retrieval and generation stages [7].

LIME (Local Interpretable Model-agnostic Explanations) is an approximation of complex models at a local scale by training a simple linear model on perturbed text samples. It

determines the words that contribute the most to a prediction providing rapid and intuitive information applicable in real-time clinical settings.

The SHAP (SHapley Additive exPlanations) is based on cooperative game theory, giving each feature a contribution score to the output of the model, and is used to estimate how much each attribute alters predictions compared to a baseline.

SHAP in XAI-RAG explains why this or that evidence was retrieved, and LIME shows how the obtained evidence affected the response that was generated—as one chain of interpretable reasoning.

C. Advancements in AI for Medical Informatics

Medical informatics based on AI has demonstrated a high level in the complexity of data analysis: CT scans, MRIs, EHRs, and in many cases with more accuracy than human specialists in identifying pathologies and risks of disease. Large language models (LLMs) can now apply this functionality to administrative and research problems, including, but not limited to summarizing clinical notes, extracting structured data, and synthesizing scientific literature.

III. THE XAI-RAG FRAMEWORK: ARCHITECTURE AND IMPLEMENTATION

Our XAI-RAG framework is a multi-component system that is aimed at the transparent and comparative analysis of medical reports. It consists of an architecture that incorporates strong data ingestion, empirical model selection, a dual-LLM generation core, and an end-to-end explainability pipeline. The entire flowchart of the methodology is presented in Fig. 1.

A. Implementation Details

The framework was implemented with specific hyperparameters to ensure reproducibility and optimal performance. The computational environment consisted of a server equipped with a Tesla T4 GPU with 15GB of VRAM. The local T5-Flan model required approximately 6GB of VRAM. The average processing time per query was approximately 2 seconds for T5-Flan and 1.5 seconds for the GROQ-GPT API. Key parameters are detailed in Table I.

TABLE I: Hyperparameters and Configuration.

Component	Parameter Value
T5-Flan	google/flan-t5-large, max_tokens=150, temperature=0.0
GROQ-GPT	openai/gpt-oss-120b, max_tokens=200, temperature=0.0
Embeddings	chunk_size=300, chunk_overlap=50
Retrieval	top_k=5, similarity_threshold=0.8

The study adopts an automated evaluation scheme that performs systematically test of both T5-Flan and GROQ-GPT to a clinical corpus. Two different prompting strategies are stipulated:

- **strict_extraction:** “Posing the task as a simple information retrieval, and telling the model to extract the exact answer to the following question: {question}”

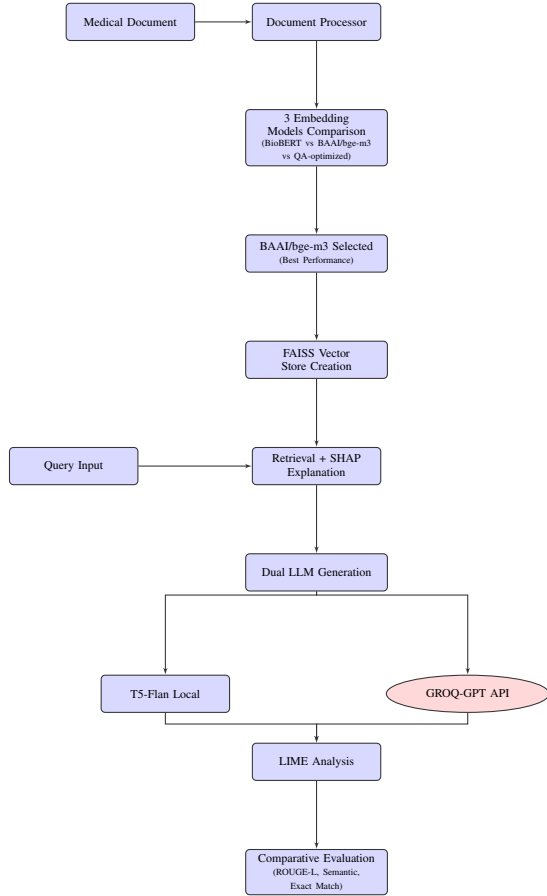


Fig. 1: Methodology flowchart of the XAI-RAG framework depicting the entire pipeline of document processing to embedding model comparison, creating the vector store, retrieving with SHAP explanations, generating dual LLM, and comparing it with the LIME analysis.

- **comprehensive_analysis:** “You are a board-certified radiologist. Analyze the provided clinical evidence and generate a concise yet comprehensive medical interpretation for the following question. Keep in mind that you are to incorporate appropriate findings, logic, and clinical implications on the basis of evidence only: {question}”

Both strategies are implemented on both models on every clinical query thus delivering four outputs.

B. Vectorization Strategy: An Empirical Analysis of Embedding Models

The model used to embed is a very important architectural decision in an RAG system because it directly will affect the quality of the semantic search and retrieval. Instead of guessing the superiority of domain-specific models, the framework takes an empirical and data-driven model selection strategy, managed by the EmbeddingComparator class. This approach questions the common wisdom that using a model pre-trained on a domain-specific corpus is always the best choice, a notion challenged by other retrieval strategies [5]. The comparative

TABLE II: Embedding Model Comparison by Derived Similarity Score. Higher derived similarity scores indicate better performance.

Query	BioBERT	General	QA Optimized
What are the findings in the brain scan?	0.0310	0.5144	0.0381
Are there any fractures detected?	0.0280	0.5527	0.0355
What is the condition of the ventricular system?	0.0358	0.5277	0.0354
What does the impression suggest?	0.0221	0.5001	0.0328

analysis of three different classes of embedding models was performed and it is shown in Table II.

- 1) `dmis-lab/biobert-base-cased-v1.1`: A pre-trained BERT-based model on large-scale biomedical corpora (Domain-Specific).
- 2) `sentence-transformers/multi-qa-mpnet-base-dot-v1`: A model fine-tuned on question-answering and semantic search tasks (Task-Optimized).
- 3) `BAAI/bge-m3`: A large general-purpose model with state-of-the-art, multi-lingual, multi-granular capability (Generalist Approach).

The findings of this empirical study were categorical. BAAI/bge-m3 model had the best average Derived Similarity Scores among all the test queries, showing better retrieval performance. This counter-intuitive result indicates that, in the task of retrieving relevant and context-rich passages of clinical reports, the architectural superiority of a large-scale generalist model supersedes the vocabulary-specific benefits of a domain-trained model such as BioBERT, a finding consistent with other recent comparisons [12]. This evidence-based fact resulted in the choice of BAAI/bge-m3 as the main embedding model of the XAI-RAG framework.

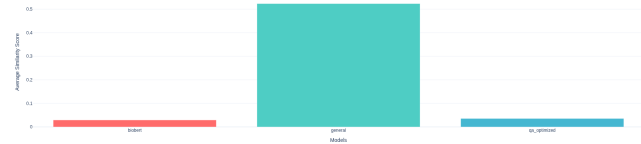


Fig. 2: Embedding Model Performance Comparison. This chart provides visual evidence supporting the selection of BAAI/bge-m3 as the superior model for the retrieval task.

C. The Dual-LLM Core: Integrating Local and Cloud-Based Generation

The key innovation in the architecture of the framework is the dual-LLM generation core at its heart, which is intended to capture a fundamental duality in clinical information requirements. The workflow of a clinician needs to involve not only the exact, factual information gathering and synthesis, global analysis. We deliberately designed our architecture to support and contrast models that have been optimized respectively to each of these requirements. The two-model system is coordinated by a combined CombinedLLMManager.

1) *Local Inference with T5-Flan*: The framework has a local inference component which is controlled by the T5FlanManager class and is constructed using the google/flan-t5-large model. This model, with 780 million parameters, was chosen because it performed well on instruction-following tasks, and because it was efficient and could be deployed on-premise. The model is automatically transferred to a CUDA-enabled GPU to increase inference speed. This local component is the route that healthcare institutions follow to prioritize data privacy and cost predictability.

2) *High-Performance API Inference with GROQ-GPT*: The cloud component is handled by the GroqGPTManager class and allows access to the openai/gpt-oss-120b model, a powerful 120-billion-parameter open-weight language model. It can be accessed through the Groq API, which is characterized by fast inference speed, enabled by a custom-built Language Processing Unit (LPU) hardware stack. The manager does secure authentication with the Groq service and offers an interface to send prompts and receive generations. Parameters of the generation are configured to achieve high-fidelity, deterministic output that would be applicable in clinical fact-finding. This API-based component is the gateway to the provision of institutions that want to take advantage of state-of-the-art reasoning and generation facilities.

D. The End-to-End Explainable Pipeline

The main object that implements the “glass box” principle is the ExplainableRAGPipeline class. It combines the chosen embedding model, the dual-LLM generation core, and the set of XAI tools into one transparent workflow.

1) *Query Deconstruction: Entity Recognition and Semantic Expansion*: Transparency starts by knowing how the system perceives the intent of the user. When a query is received, it is run through a SpaCy NLP model and linguistically analyzed. This consists of Named Entity Recognition (NER) to recognise relevant medical or anatomical terms, and dependency parsing to learn the grammatical structure of the question. The system also does query expansion to make the successive retrieval phase more robust. It expands the query with terms related to a given term in a pre-defined dictionary of medical synonyms. All this information, the original query, the identified entities, the key terms and the expanded terms are then presented to the user producing a clear description of how their query was treated.

2) *Transparent Retrieval: Explaining Relevance with Similarity and SHAP*: When the query is processed, the system extracts the most pertinent text chunks out of the vector store. The explain_retrieval procedure aims at ensuring that this step is completely transparent. On every piece of chunk retrieved, it gives a multiple-layered explanation. The raw similarity score, a preview of the content, and its length are provided as the first layer. The second level offers heuristic-based relevance description, indicating the extent of term overlap between the most important terms in the query and the terms in the chunk. The third level of explanation is based on SHAP (SHapley Additive exPlanations). A simplified SHAP

analysis is conducted to present a feature-based attribution of the retrieval decision, treating the similarity prediction of the retrieval model as the outcome. SHAP can be used to discover which specific words or features in the text chunk may have contributed most to the high similarity score. This goes beyond a mere score to address the vital question as to why a given piece of evidence was considered significant by the system.

3) *Interpretable Generation: Auditing LLM Responses with LIME*: The last pipeline step is the generation, during which the LLM summarizes the evidence retrieved into a coherent response. The explain_generation approach gives a comparative discussion on the results of both T5-Flan and GROQ-GPT. The quantitative measures presented in this analysis are the length of a response and the frequency of medical words, and the qualitative measures are the occurrence of words indicating high, medium, or low confidence.

To give a more in-depth explanation into the process of generation itself, the framework uses LIME (Local Interpretable Model-agnostic Explanations) [8], [9]. To examine the impact of the final augmented prompt on LLM output, LIME is utilised. LIME constructs a local linear model by perturbing the input prompt and maps the corresponding impact on the answer generated, determining which words in the input prompt were the most important to determining the output answer. This enables a user to confirm that the answer that the LLM actually gives is based on the evidence that was given in the context.

E. Advanced Structural Explanations: Knowledge Graph Construction and Centrality Analysis

To give a more comprehensive interpretation of the information contained in the source documents, the framework comes with an AdvancedXAIAnalyzer module, which can build and analyze a medical knowledge graph. A knowledge graph is a model of information in the form of a network of entities (nodes) and their relationships (edges), which can yield insight into underlying patterns and relationships that are not obvious in textual form [10], [11].

The building process starts with the application of SpaCy NLP model to process the data of clinical texts. The named entities recognized, e.g. the concept “Brain”, “Ventricular system”, or a particular medical finding, are inserted as nodes in a directed graph. Dependency parsing is then utilized in the inference of the relationships between these entities. The unstructured text of the medical reports is converted into a structured, machine readable, graph of medical concepts and relationships.

After the knowledge graph is built up, the framework uses graph-theoretic algorithms to examine the structure and find the most significant concepts. It computes three centrality measures:

- 1) **Degree Centrality**: Determines nodes with the greatest number of direct connections representing concepts that tend to be mentioned most together with others.

- 2) **PageRank:** Determines nodes that are pointed to by other significant nodes, which are the influential concepts in the document.
- 3) **Betweenness Centrality:** Reveals nodes, which are often part of the shortest paths among other nodes, and thus important as bridge concepts.

IV. EXPERIMENTAL DESIGN AND EVALUATION PROTOCOL

A. Dataset and Ground Truth Corpus

The main dataset of all the experiments is one representative medical document CTScanExt.docx. This report represents a C.T. scan of the brain that is a de-identified report document; it has sections namely technique, findings and impression. It contains a combination of structured observation, numeric values, and descriptive radiology terms.

A high quality ground truth corpus was developed using a scalable approach. There were a set of five canonical clinical questions developed and they comprised of the most important findings and information in the source document. The following questions were then asked to Grok AI, a state-of-the-art general purpose frontier model. The answers produced by this model were detailed, well organized, and clinically coherent and were gathered and presented in a different file, CT Scan Case.docx. This is the document used as the ground truth or reference standard to all subsequent evaluations.

B. The “Chain of Trust” Evaluation Metrics: ROUGE-L, Semantic Similarity, and Exact Match

Medical text generation cannot be measured based on a single time, because a high lexical similarity does not imply the same fidelity to semantics or factual accuracy. To resolve this, the framework uses a “Chain-Of-Trust” protocol, which assembles 3 complementary metrics, each focusing on a different factor of answer quality. The development of robust evaluation frameworks is a known challenge in RAG pipelines [3].

- **ROUGE-L (Longest Common Subsequence):** This is a metric that quantifies the overlap of words and phrases between generated and reference answers, and therefore captures key terminology and word order. It acts as a Keyword Checker.
- **Semantic Similarity:** This metric is based on computing cosine similarity between answers embedding and assesses the conceptual similarity. It serves as the “Meaning Judge,” which identifies lexically different but identical clinical paraphrases.
- **Exact Match:** This is the strictest one and it compares characters and scores with a binary value. As the “Strict Grader,” it is essential to confirm the accuracy of specific data like indices, identifiers, and medical codes.

V. RESULTS AND DISCUSSION

A. Embedding Model Performance

The general purpose BAAI/bge-m3 exceeded both BioBERT and multi-qa-mpnet in retrieval, contrary to expectation. Although BioBERT specialised vocabulary

is valuable in fine-grained tasks such as Named Entity Recognition, retrieval tasks demand strong processing of long biomedical text. The architectural advantages of bge-m3 namely its higher parameter count, wider training corpus and longer training context length (8192 tokens) were more decisive. This indicates that for RAG tasks, architectural sophistication can be more important than domain-specific pretraining.

B. Comparative Model Evaluation: T5-Flan vs. GROQ-GPT

The overall assessment reveals a set of obvious disparities: the GROQ-GPT is slightly superior in ROUGE-L (lexical overlap) and is lower in Semantic Similarity. Both did not pass on the Exact Match, as is the descriptive nature of clinical reporting. Depending on which aspect is most important either lexical recall or semantic fidelity, the decision is made based on which of the two is more important as in Fig. 3 and Table III.

TABLE III: Aggregate Dual-LLM Evaluation Results. GROQ-GPT outperforms on lexical ROUGE-L, whereas T5-Flan is stronger on Semantic Similarity.

RAG Approach	ROUGE-L	Sem. Sim.	Exact Match
<i>T5-Flan</i>			
Strict Extraction	0.196	0.440	0.000
Comp. Analysis	0.196	0.418	0.000
Overall Avg.	0.196	0.429	0.000
<i>GROQ-GPT</i>			
Strict Extraction	0.270	0.418	0.000
Comp. Analysis	0.209	0.403	0.000
Overall Avg.	0.239	0.410	0.000

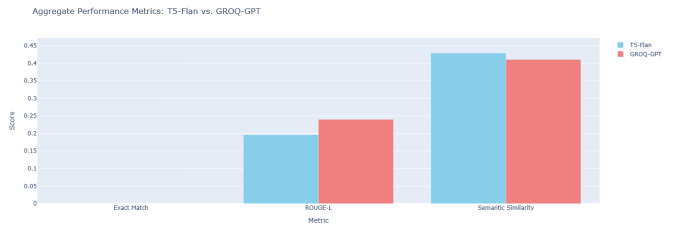


Fig. 3: Aggregate Performance Metrics of T5-Flan vs. GROQ-GPT. Visual summary of trade-offs between lexical and semantic performance.

C. RAG Approaches: Extraction vs. Analysis

Task framing was also a moderator of model performance. T5-Flan outperformed semantic fidelity as intended, because of its advantages in instruction adherence, and GROQ-GPT achieved higher lexical overlap. The generative capacity of GROQ-GPT generated a more detailed and context-integrated response in comprehensive analysis. Structured extraction tasks are more appropriate to T5-Flan, and interpretive tasks are better leveraged by GROQ-GPT, as in Fig. 4.



Fig. 4: Performance by RAG Approach: T5-Flan vs. GROQ-GPT. Comparison of extraction vs. analysis framing.

D. Explainability Insights from LIME

Qualitative validation was done using explainability tools as demonstrated in Fig. 5. Query terms having LIME score of 0.0 indicated that evidence was the only factor that contributed, which reduced risk of hallucination. This prompted a Query Influence Score: low = evidence-based, high = over-reliance on prompt. Clinically significant entities were also highlighted with knowledge graph centrality and thus were interpretable.

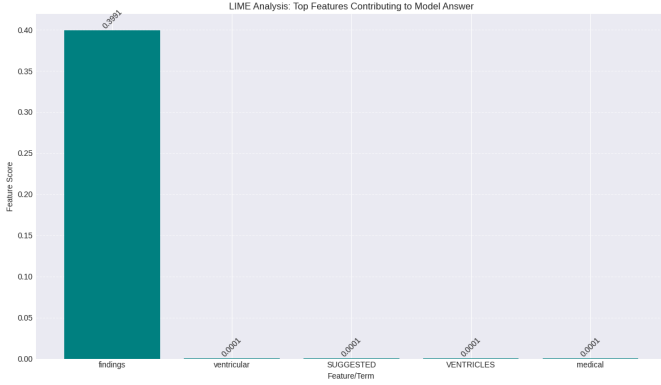


Fig. 5: LIME Analysis: Most influential words from the prompt in shaping the model's response.

VI. CONCLUSION

This research presents a transparent, dual-LLM XAI-RAG model to solve the major issue of explainability in clinical AI. With the XAI tools such as LIME and SHAP built directly into the RAG pipeline, we have now a glass box system in which the logic of the whole process: query interpretation to the evidence based generation of an answer, can be audited. Our empirical assessment performance indicated a major trade-off between a local T5-Flan model that has high semantic fidelity with a high-performance GROQ-GPT model that has high lexical accuracy. The demonstration that a general-purpose embedding model can be more competent than a domain-specific model is an indicator of the significance of architectural sophistication in retrieval. Such a framework provides a viable way of getting credible AI tools that can streamline clinical processes by offering valuable insights that are reliable, interpretable, and verifiable to complex medical data.

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